

Final Report Template

Hotel Booking Demand Clustering Study

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Abstract

This study investigates whether latent behavioural segments exist among hotel bookings made through Travel Agency/Tour Operator (TA/TO) intermediaries, using a publicly available dataset of 97,870 records from two Portuguese hotels (Algarve and Lisbon, 2015–2017). Bookings are represented as a 57-dimensional mixed-type feature matrix (R0), combining StandardScaler-normalised numerical variables with one-hot encoded categoricals; all features are measured at booking time to avoid leakage. Three clustering families are applied and compared: K-Means, iK-Means (AP initialisation), and Ward hierarchical clustering, evaluated through internal validity indices (Silhouette, Calinski-Harabász, Davies-Bouldin), multi-seed stability analysis, cross-method Adjusted Rand Index, and a RobustScaler sensitivity variant. K-Means at $K = 4$ achieves tied-best internal indices (Silhouette = 0.140) with near-perfect seed stability (ARI = 0.995) and cross-method agreement with iK-Means (ARI = 0.994), yielding four interpretable profiles: committed resort families with parking (C0), mainstream online transient city bookers (C1), offline long-lead service-heavy bookers (C2), and speculative non-refundable group blocks with 99% cancellation (C3). The main limitation is the absence of ground-truth labels, meaning validity remains internal and post-hoc; results are further constrained to TA/TO bookings from two Portuguese properties between 2015 and 2017.

Keywords: hotel booking demand; clustering; segmentation; mixed-type data; reproducibility

1. Introduction

The global hospitality industry relies increasingly on intermediaries, travel agencies and tour operators (TA/TO), to fill hotel capacity across both leisure and city markets. Within this intermediary channel, demand is far from monolithic: a long-haul package tourist booked months in advance through an online platform occupies a fundamentally different commercial position than a last-minute group reservation funnelled through an offline operator. Yet hotels seldom have class labels that would allow a supervised model to distinguish such profiles. The only observational record available at the moment a booking is confirmed consists of the booking itself, its timing, composition, stay structure, and the commercial metadata attached by the intermediary.

The segmentation question addressed in this report is formalised as RQ1: Among bookings made through Travel Agency / Tour Operator (TA/TO) intermediaries, are there latent behavioural groups characterised jointly by booking timing, stay design, party composition and commercial-channel metadata that are observable at booking time? The question is deliberately grounded in a single, well-defined sub-population, the 97,870 TA/TO records drawn from a publicly available hotel booking demand dataset (António, de Almeida, & Nunes, 2019) covering arrivals at a resort hotel in the Algarve and a city hotel in Lisbon between July 2015 and August 2017.

Similarity between bookings is operationalised through Euclidean distance on a mixed numerical-categorical feature matrix (representation R0). This matrix stacks eight StandardScaler-normalised numerical variables, lead time, total nights, weekend share, party size, previous cancellations, previous non-cancelled bookings, parking spaces requested, and special requests, one binary indicator (repeated guest), and a one-hot-encoded categorical block covering market segment, reserved room type, meal plan, customer type, deposit type, country of origin, and arrival month. All features are measured at booking time; outcome and post-arrival variables such as cancellation status, booking changes, and assigned room type are explicitly excluded from the clustering inputs and reserved for post-hoc profiling only. Under this definition, two bookings are similar if they share booking-horizon behaviour, stay configuration, group composition, and commercial positioning, a notion of proximity that maps directly onto actionable guest archetypes.

The main contribution of this report is a structured, reproducible pipeline that moves from data preparation through clustering, evaluation, and substantive interpretation. On the methodological side, three clustering families are applied to the same representation: K-Means, Agglomerative Hierarchical Clustering with Ward, and Fuzzy c-Means. These are evaluated through a unified protocol, combining internal validity indices, such as Silhouette, Calinski-Harabasz, and Davies-Bouldin, with multi-seed stability analysis, cross-method Adjusted Rand Index comparison, and a controlled preprocessing variant (RobustScaler as R1). Two additional extension analyses enrich the picture: a cluster-aware anomaly detection study that identifies the most atypical TA/TO bookings relative to the learned partition, and a PCA/SVD representation study that empirically tests whether linear dimensionality reduction preserves or degrades the cluster structure found in the original feature space. The final segmentation yields interpretable booking profiles that can directly inform revenue management, capacity planning, and intermediary-specific marketing strategies.

2. Dataset and Preprocessing

2.1 Dataset documentation

The analysis is based on the Hotel Booking Demand dataset, a publicly available record of hotel reservations originally published by António, de Almeida, and Nunes (2019) and distributed in this course via the official course release. The dataset covers two Portuguese properties; a resort hotel located in the Algarve (H1) and a city hotel located in Lisbon (H2) and records all bookings with scheduled arrival dates between 1 July 2015 and 31 August 2017. Each row represents one individual booking record, with 32 attribute columns spanning guest composition, stay structure, booking channel, commercial metadata, and operational outcomes. The table below documents the dataset in a reproducible and traceable form.

Table 1| Overview of the dataset used, including source, size, time span, and reproducibility conditions.

Item	Detail
Dataset and version	Hotel Booking Demand; course release (hotel_bookings.csv). Reference: António, N., de Almeida, A., & Nunes, L. (2019). <i>Hotel booking demand datasets</i> . <i>Data in Brief</i> , 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126
Unit of analysis	One row = one booking record, corresponding to a single reservation at one of the two hotels.
Size and time span	119,390 rows × 32 columns. Arrivals span 1 July 2015 – 31 August 2017 across H1 (Resort Hotel, Algarve) and H2 (City Hotel, Lisbon).
Source and terms	Course-distributed release of the dataset described in António et al. (2019). License: Creative Commons Attribution 4.0 International (CC BY 4.0). No usage restrictions beyond attribution.
Checksum / hash	MD5 hash of the raw CSV computed programmatically at runtime via hashlib.md5 and stored in DATASET_MD5. All six notebooks verify against the same file at load time, ensuring identical input across the full pipeline.
Raw-data handling	The raw CSV is not committed to the repository. To reproduce results, place hotel_bookings.csv in data/raw/relative to the project root. All notebooks resolve the path dynamically via <code>_find_project_root()</code> and raise a <code>FileNotFoundError</code> with explicit placement instructions if the file is absent.

2.2 Data quality and feature governance

- **Missingness and imputation:**

The missingness audit was performed on the TA/TO sub-population, and only two columns contain genuine missing values. Country is missing for a small share of records; following the dataset reference paper, which notes that absent country typically corresponds to the resort's domestic market, missing values are imputed as "PRT". Children contain four NaN rows, imputed as 0 on the grounds that an absent value for a count field plausibly represents zero children.

- **Outlier handling and record exclusions:**

- Zero-guest records. Bookings where adults + children + babies == 0 are dropped as they do not represent a valid behavioural unit. This affects a handful of records;
- ADR anomalies. Records with $\text{adr} \leq 0$ (complimentary or test entries) and the single record with $\text{adr} \approx 5,400$ (a clear data-entry error) are filtered from the ADR-clean subset (used exclusively for post-hoc profiling and the with ADR sensitivity variant);
- Lead time. Values exceeding 365 days are noted (a small proportion of the sub-population) but are not removed; StandardScaler normalisation mitigates their influence in the Euclidean space, and the distribution is documented in the EDA.

The final working sub-population after all exclusions contains approximately 97,870 TA/TO records.

- **Segmentation and index time:**

The segmentation time is defined as immediately after the booking is recorded. This is the operationally meaningful moment at which a hotel or revenue management system would assign a new booking to a segment. Any variable whose value can change after this moment is excluded from the clustering inputs to avoid leakage. The excluded post-event variables are `is_canceled`, `reservation_status`, `reservation_status_date`, `booking_changes`, `assigned_room_type`, and `days_in_waiting_list`. These variables are retained solely for post-hoc cluster profiling.

- **High-cardinality and ID-like fields:**

Agent and company are excluded from clustering entirely. Within the TA/TO sub-population agent alone takes over 3,000 distinct values, including it would cause clusters to reflect agency identity rather than booking behaviour. Company is similarly excluded due to high cardinality and extreme sparsity. For the categorical inputs that are retained, levels appearing in fewer than 1% of the sub-population records are collapsed into an "Other" category before one-hot encoding. This prevents rare dummies from distorting Euclidean distances while keeping the dominant levels individually interpretable.

Additionally, `arrival_date_year` is excluded because it spans only three calendar years (2015–2017), meaning any year-based split would reflect the dataset's temporal extent rather than genuine behavioural differences. `arrival_date_week_number` and `arrival_date_day_of_month` are excluded for redundancy and noise reasons, with seasonality captured at month granularity via `arrival_date_month`. The raw stay-night and guest-count columns (`stays_in_week_nights`, `stays_in_weekend_nights`, `adults`, `children`, `babies`) are replaced by three engineered features, `total_nights`, `weekend_share`, and `party_size`, that summarise the same information more compactly and at the appropriate behavioural level.

- **Final representation and implied distance metric**

The final representation, labelled R0, combines three blocks after the sub-population filter and all cleaning steps:

Table 2| Selected variables and their encoding methods, organised by type (numerical, binary, and categorical).

Block	Variable	Encoding
Numerical	<code>lead_time</code> , <code>total_nights</code> , <code>weekend_share</code> , <code>party_size</code> , <code>previous_cancellations</code> , <code>previous_bookings_not_canceled</code> , <code>required_car_parking_spaces</code> , <code>total_of_special_requests</code>	StandardScaler (zero mean, unit variance). No log transform in R0
Binary	<code>is_repeated_guest</code>	Retained as 0/1 without rescaling, to mirror the one-hot block
Category	<code>market_segment</code> , <code>reserved_room_type</code> , <code>meal</code> , <code>customer_type</code> , <code>deposit_type</code> , <code>country</code> , <code>arrival_date_month</code>	OneHotEncoder after rare-category grouping (1% threshold); dummies not variance-standardised

The resulting matrix has the shape $n \times p$, where $n \approx 97,870$ and p reflects the stacked numerical, binary, and one-hot columns. The implied similarity measure is Euclidean distance computed directly on this matrix. Because numerical variables are standardised and one-hot columns are kept as 0/1, the Euclidean metric assigns comparable weight to continuous behavioural dimensions and categorical membership indicators. `adr` is deliberately excluded from R0: as a revenue outcome partially determined by the channel and room type, its inclusion could cause clusters to reflect pricing tiers rather than booking behaviour. It is used exclusively in post-hoc profiling and in a controlled sensitivity variant (R2).

3. Clustering Methods

To extract robust customer booking profiles from the Travel Agency/Tour Operator (TA/TO) sub-population, three distinct clustering methodologies and one specialized initialization pipeline are deployed. All models operate within the exact same Euclidean feature representation space (R0).

All methods are applied to the same data representation (R0) and the same sub-population (TA/TO bookings, $n \approx 97,840$ after cleaning), ensuring that any difference in cluster quality is attributable to the algorithm and not to preprocessing choices. Euclidean distance is computed on the final encoded matrix X_{R0} , which stacks StandardScaler-scaled numerical variables (no log transform) with one-hot encoded categorical variables (rare categories grouped under "Other" at a 1% minimum-frequency threshold). One-hot columns are kept as {0, 1} and are not variance-standardised. ADR is excluded from the clustering representation; it is used only in post-hoc profiling.

Clustering inputs (numerical, StandardScaler): `lead_time`, `previous_cancellations`, `previous_bookings_not_canceled`, `total_nights`, `weekend_share`, `party_size`, `required_car_parking_spaces`, `total_of_special_requests`.

Binary input (unscaled): `is_repeated_guest`.

Categorical inputs (one-hot): `market_segment`, `reserved_room_type`, `meal`, `customer_type`, `deposit_type`, `country`, `arrival_date_month`.

Post-hoc / profiling-only (excluded from clustering): `is_canceled`, `reservation_status`, `hotel`, `booking_changes`, `adr`, `assigned_room_type`.

3.1 Baseline Family – K-Means and iK-Means

The primary baseline model is standard **K-Means** clustering. It is an iterative partition-based algorithm that divides n observations into K disjoint clusters by minimizing the within-cluster sum of squares (inertia):

$$J = \sum_{j=1}^K \sum_{n \in S_j} |x_n - \mu_j|^2, \quad (1)$$

K-grid: $k \in \{2, 3, 4, 5, 6, 7, 8\}$: predeclared before any model is fitted. The lower bound ($k = 2$) ensures a non-trivial partition; the upper bound ($k = 8$) follows the project brief recommendation and the domain plausibility argument that more than eight operationally distinct TA/TO booking archetypes cannot be meaningfully actioned.

k-selection rule (declared before running): choose the k that maximises the Silhouette score and is at, or just past, the elbow inflection of the inertia curve. If the Silhouette argmax and the elbow disagree, prefer the smaller k (parsimony) provided the Silhouette gap between the two candidates is < 0.02 . Calinski-Harabasz (CH) and Davies-Bouldin (DB) are reported as supporting evidence only.

Selected $k = 4$: the Silhouette across $k \in \{4, 5, 6, 7, 8\}$ differs by less than 0.005 (max at $k = 8$ with Sil = 0.1425; $k = 4 = 0.1403$); the parsimony rule therefore selects $k = 4$, confirmed by the elbow inflection of the inertia curve. Full results across the k -grid are reported in Appendix Table A1.

iK-Means is Mirkin's Iterative Anomalous-Pattern (AP) procedure. It produces a deterministic set of initial centroids by iteratively extracting "anomalous clusters" from the residual data. At each iteration: (1) the entity farthest from the grand mean in the range-standardised space is selected as a tentative seed; (2) an AP cluster is grown by alternating assignment (observations strictly closer to the tentative centroid than to the grand mean) and centroid recomputation; (3) convergence is declared when membership stops changing (exact stability criterion) or centroid displacement $\leq \epsilon = 10^{-9}$; (4) the cluster's indices are removed from the residual set. The procedure terminates when the residual set is empty.

Because the AP procedure scales as $O(n^2)$ in distance computations, it is run on a fixed seeded subsample of 20,000 records (seed = 12345); the resulting centroids are then propagated to the full sub-population.

Selected $k = 4$: (identical to K-Means, enabling a direct head-to-head comparison via ARI). Detailed parameter tables are provided in Appendix Table A2.

3.2 Alternative Family – Agglomerative Hierarchical Clustering (Ward Linkage)

Ward linkage is conceptually distinct from the centroid-update family: it is hierarchical, fully deterministic, and uses a divisive-free agglomerative procedure that at each step merges the pair of clusters whose union minimises the increase in total within-cluster variance (WCSS).

Linkage: Ward. Distance: Euclidean (required by Ward's WCSS criterion).

k-range: $k \in \{2, 3, 4, 5, 6\}$: the upper bound is tighter than for K-Means for domain plausibility: a hierarchical cut at more than six clusters produces groups too small to be operationally meaningful, and the dendrogram provides direct guidance on natural cut points.

Scalability: agglomerative Clustering requires $O(n^2)$ memory for the full linkage matrix, which is infeasible at $n \approx 97,840$. The dendrogram and Ward fit are therefore computed on a fixed subsample of 10,000 rows (seed = 12345, same subsample used for the dendrogram diagnostic). Each full-dataset point is then assigned to its nearest Ward centroid (Euclidean nearest-centroid propagation). This is the standard practical compromise for large- n hierarchical clustering and is reported transparently in the experiment log.

k-selection rule: silhouette argmax on $k \in \{2, \dots, 6\}$ (primary), confirmed by the merge-distance profile from the dendrogram (largest upward gap $\rightarrow$ natural cut point), with a parsimony tiebreak (smallest k within 0.005 of the Silhouette maximum).

Selected $k = 2$. The Silhouette argmax pointed to $k = 2$, confirmed by the family-specific dendrogram diagnostic. Detailed parameter tables are in Appendix Table A3.

Family-specific justification for the alternative: Ward hierarchical clustering is conceptually distinct from K-Means along three axes: (i) it is deterministic and produces a full merge history rather than a single flat partition; (ii) it does not use iterative centroid updates; and (iii) it provides a dendrogram that allows the hierarchical structure of booking behaviour to be inspected visually, offering qualitative insight unavailable from centroid-based methods. These properties make it a genuinely complementary view rather than a variant of the same computational family.

3.3 Fuzzy c-Means

Fuzzy c-Means (FCM) is conceptually distinct from both the centroid-update partition family and the agglomerative hierarchical family: it is a soft-clustering algorithm where data points are not restricted to a single rigid group. Instead, each booking maintains a continuous membership degree across all clusters simultaneously, subject to the probabilistic constraint that the sum of memberships across all clusters equals one. This approach allows the model to capture overlapping behavioral profiles and diffuse boundaries between segments.

Objective Function and Parameters: the model minimizes a fuzzified version of the within-cluster sum of squares: where the fuzziness weighting exponent is. The fuzziness index is Euclidean Distance (applied over the main representation).

k-range: this window matches the range analyzed in the alternative hierarchical family to ensure a synchronized comparative baseline across both advanced modeling structures.

Scalability and Implementation: The FCM algorithm was coded from scratch using an Alternating Optimization (AO) loop that updates the membership matrix and the prototype matrix iteratively.

k-selection rule: hardened Silhouette argmax on (primary), where soft memberships are converted to hard labels via an argmax operator () for direct metric alignment, complemented by specialized soft partition indices: the Partition Coefficient (argmax closer to 1 implies cleaner separation) and Classification Entropy (lower values indicate higher assignment confidence).

Selected k = 4: the hardened Silhouette argmax points structurally to $k = 2$ (consistent with Ward's natural split), but $k = 4$ is adopted to enable a direct head-to-head ARI comparison with the K-Means / iK-Means partition and to retain the four behaviorally distinct profiles that motivate the study. The fuzzifier m is calibrated on a $\{1.05, 1.08, 1.10, 1.13, 1.15, 1.20, 1.50, 2.00\}$ grid (Appendix Table A8); $m = 1.13$ is selected as the smallest value yielding a "soft but informative" partition (Partition Coefficient = 0.853, mean max membership = 0.898, ARI vs. K-Means = 0.827) before the membership matrix degenerates to uniform at $m \geq 1.50$.

4. Experimental Protocol

4.1 Finess Core Setup and Predefined Parameters

A strict grid of candidate clusters was predefined and swept uniformly across K-Means, iK-Means, and Ward Hierarchical Clustering to ensure a common baseline for comparison.

For the Fuzzy c-Means model, a structural grid search was conducted over α . This sweep aimed to identify the exact tipping point where the membership transition shifts from highly informative to a degenerate uniform distribution ($\alpha = 0$). The empirical search selected a tight exponent of $\alpha = 0.5$, operating safely within the informative domain while allowing slight boundary adjustments.

4.2 Fairness and reproducibility

To isolate models from random convergence anomalies, a robust stability verification framework was integrated:

- **K-Means & iK-Means Stability:** evaluated using a sequence of 10 distinct, predefined pseudo-random seeds: [12345, 23456, 34567, 45678, 56789, 67890, 78901, 89012, 90123, 11111]. We report the mean and standard deviation across these 10 runs for all validity indices, alongside the mean pairwise Adjusted Rand Index (ARI) to check label consistency across different initializations;
- **Ward Hierarchical Stability:** because Ward linkage is fundamentally deterministic, standard seed variance does not apply. Instead, a bootstrap resampling protocol is introduced. The model is repeatedly fitted on 5 independent bootstrap samples ($n=10,000$ observations drawn with replacement), and stable metrics are tracked across the resamples.

4.3 Validity indices and variability reporting

- **Silhouette score (primary):** measures mean intra-cluster cohesion minus inter-cluster separation, normalised to $[-1, 1]$; higher is better. Computed on a fixed seeded subsample of 10,000 rows ($O(n^2)$ safeguard). This is the primary selection criterion;
- **Calinski-Harabasz index (supporting):** ratio of between-cluster to within-cluster dispersion; higher is better. $O(n)$, computed on the full sub-population;
- **Davies-Bouldin index (supporting):** average similarity ratio of each cluster to its most similar neighbour; lower is better. $O(n)$, computed on the full sub-population;
- **Fuzzy Validity Sub-Suite:** for FCM, specialized soft-partition metrics are included to monitor membership quality, *Partition Coefficient*, that racks crispness where higher values mean a cleaner separation, *Classification Entropy*, measures uncertainty, and lower values indicate higher assignment confidence, and *Xie-Beni Index*, where a ratio of total fuzzy compact variance to minimal prototype distance; lower scores indicate well-separated prototypes.

4.4 Validity family specific justification for the alternative clustering

Ward linkage minimises total within-cluster variance at each merge step, the same WCSS criterion as K-Means but algorithmically distinct (hierarchical, deterministic, no centroid updates). The dendrogram it produces is a family-specific diagnostic unavailable from centroid-based methods: the merge-distance profile identifies the most natural number of clusters by locating the largest upward jump in WCSS (the "gap"), and the truncated

dendrogram provides a visual record of the merging hierarchy. This was used to confirm the $k = 2$ selection independently of the Silhouette index.

4.5 Controlled sensitivity and preprocessing variant

To ensure the identified clusters are structurally stable and not artifacts of specific preprocessing choices, two controlled sensitivity experiments were executed:

- **Outlier Scaling Variant (R1 - RobustScaler Sensitivity):** the entire baseline configuration is mirrored, but StandardScaler is replaced with a localized RobustScaler (centering by median and scaling via the Interquartile Range). This tests whether extreme high-variance attributes (such as long lead_time values) pull cluster centroids artificially;
- **Linear Dimensionality Reduction Check (Module E4 - PCA Space Study):** the core R0feature space is projected into a lower-dimensional subspace via Principal Component Analysis (PCA), picking a heuristic threshold that retains 90% of the Explained Variance Ratio (EVR). K-Means is then executed in this reduced space to determine whether projecting the data preserves, improves, or degrades the cluster partitions found in the original space. Detailed parameter tables are in *Appendix Table A4*

4.6 Stability Analysis

Table 3| Perturbation protocols applied to each clustering method for stability assessment.

Method	Protocol	Perturbation type	Runs
K-Means	10 random seeds from SEEDS_STABILITY	Initialisation randomness	10
iK-Means	10 seeds, different AP subsample per seed	AP subsample + initialisation	10
Ward	5 bootstrap subsamples (90% without replacement), different seeds	Fitting-data composition	5

For each method the pairwise ARI across all 10 run-pairs is computed on the overlapping original-index set (for Ward) or on the full sub-population (for K-Means and iK-Means). The mean and standard deviation of those 10 ARI values are the reported stability statistics.

K-Means and iK-Means both show high pairwise ARI across seeds, confirming that the $k = 4$ partition is not sensitive to random initialisation. Ward, being deterministic for a fixed subsample, exhibits variability only through changes in fitting-data composition; the stability observed there measures robustness to subsample perturbation rather than algorithmic stochasticity. Detailed parameter tables are in *Appendix Table A5*.

4.7 Experimental block summary

Table 4| Experimental blocks of the study, detailing the purpose of each block and its main outputs.

Experiment block	Purpose	Main output
Baseline models	K-Means and iK-Means across $k \in \{2, \dots, 8\}$ on R0; best-k selection and profiling	Model comparison table (Appendix A1–A2) and cluster profiles
Alternative family	Ward hierarchical ($k \in \{2, \dots, 6\}$) on R0; dendrogram diagnostic; nearest-centroid propagation; comparison with K-Means via ARI	Dendrogram figure, merge-distance profile, ARI cross-tabulation
Stability and robustness checks	5 seeds/resamples per method; controlled R1 variant (RobustScaler); cross-method ARI (K-Means vs iK-Means vs Ward)	Stability summary table (mean $\pm$ std of Silhouette, CH, DB, pairwise ARI) and R0 vs R1 comparison table

5. Results and Cluster Interpretation

5.1 Data preparation outcomes

The final analytic dataset contains 97,870 rows $\times$ 57 encoded dimensions (8 numerical inputs + 49 one-hot levels). Main exclusions are non-TA/TO bookings ($\approx 18\%$), post-event variables (is_canceled, reservation_status, assigned_room_type, booking_changes, days_in_waiting_list) and the high-missing company/agent fields. Representation R0 applies StandardScaler to numerics and one-hot encoding (with 1%-minimum rare grouping) to categoricals, implying Euclidean distance over a 57-dimensional standardised vector.

5.2 Model comparison

K-Means and iK-Means both converge at K=4 with essentially identical indices; Ward selects K=2, a strictly coarser partition that hides three of the four behavioural segments described in §5.3.

Table 5| Internal indices at selected K on R0.

Method	K	Silhouette $\uparrow$	CH $\uparrow$	DB $\downarrow$	Time (s)
K-Means	4	0.140	12,956	1.921	1.02
iK-Means (AP-init)	4	0.140	12,956	1.922	1.04
Ward	2	0.318	11,961	1.151	1.14

Ward's higher Silhouette is a **granularity artefact**: K=2 isolates a tiny low-cancellation cluster (n = 5,175, 0.4%) from "everyone else" (n = 92,575, 43.3%): operationally insufficient.

Table 6| Stability across 10 seeds and 20-bootstrap resamples.

Method	K	Silhouette $\mu \pm \sigma$	Pairwise ARI (5 seeds)	Bootstrap ARI (B=20)
K-Means	4	0.139 $\pm$ 0.001	0.995 $\pm$ 0.003	0.710 $\pm$ 0.289
iK-Means (AP-init)	4	0.131 $\pm$ 0.015	0.711 $\pm$ 0.289	0.971 $\pm$ 0.015
Ward	2	0.316 $\pm$ 0.002	1.000 $\pm$ 0.000	—

Across seeds K-Means is essentially deterministic; iK-Means inherits more variance from AP initialisation. Under bootstrap resampling the picture inverts: iK-Means-AP reaches ARI = 0.971 $\pm$ 0.015 versus K-Means 0.710 $\pm$ 0.289, indicating AP-derived prototypes are more robust to *which* bookings are sampled. Cross-method ARI is K-Means-iK-Means = 0.994 and K-Means-Ward = 0.155: the two partitional methods recover the same partition, Ward a different one.

Sensitivity. PCA-21d (90% EVR) raises Silhouette to 0.157 at K=4 with ARI vs R0 = 0.93: same cluster identities, less noise. RobustScaler (R1) instead drops ARI vs R0 to 0.18 because it compresses the long-tail signal in lead_time and previous_cancellations we want to preserve; R0 is retained. Fuzzy C-Means with calibrated m = 1.13 yields PC = 0.853 and ARI vs K-Means = 0.827 ("soft but informative"); 6-8% of bookings on the C1-C2 boundary carry ambiguous memberships (max-membership < 0.6), confirming the boundary is genuinely fuzzy.

Selected model: K-Means, K=4, R0, seed 12345. Rationale: (i) tied-best internal indices with iK-Means; (ii) seed-deterministic (ARI 0.995); (iii) the partition is 99.4% identical to iK-Means, so the lower K-Means bootstrap ARI reflects boundary fuzziness rather than method failure; (iv) preserves the four behavioural segments that Ward at K=2 hides.

Figure T3.1 — Internal Validity Indices: K-Means vs iK-Means vs Ward (R0, k=best)

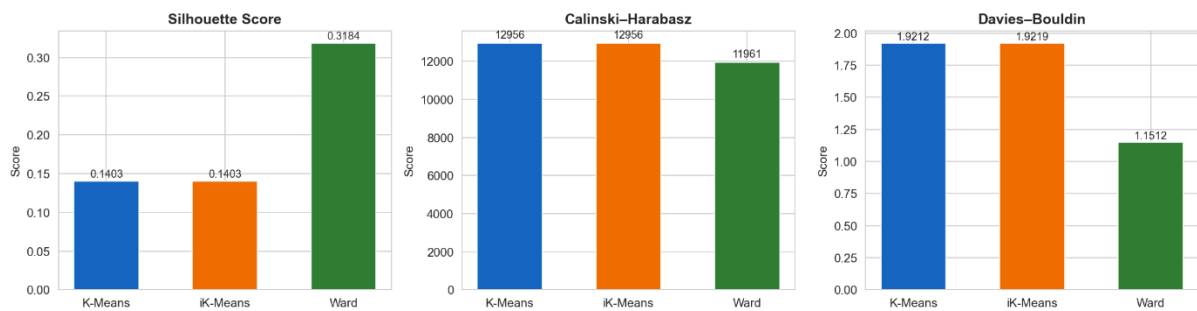


Figure 1| Internal-index curves (Silhouette, Calinski-Harabasz, Davies-Bouldin) for K-Means, iK-Means and Ward over K = 2..8 on representation R0. K-Means/iK-Means peak at K = 4; Ward peaks at K = 2 due to granularity.

5.2 Cluster profiles

Table 7| Post-hoc descriptors of the four K-Means clusters.

Cluster	N	Cancel %	City / Resort %	Median ADR (€)	Booking changes
C0	4,478	0.0	26.5 / 73.5	105.3	0.30
C1	49,498	46.1	75.2 / 24.8	89.3	0.17
C2	38,300	31.1	67.0 / 33.0	115.0	0.21
C3	5,474	99.2	87.3 / 12.7	69.0	0.03

- **C0: Parking-equipped resort families.** Parking $\approx$ +2,083% vs. grand mean (the only cluster with cars), special requests +60%, repeated-guest share +123%, lead time -46%; Online TA + No Deposit + resort-dominant (73.5%), zero cancellations, highest booking-change activity. *Committed leisure bookers, low risk.*

- **C1: Mainstream online transient, city hotel.** Largest segment (~51%): Online TA + city hotel (75%) + No Deposit; median ADR €89, 46% cancellation, lead time +12%. *Volume backbone with material cancellation hedging needed.*
- **C2: Offline TA/TO long-lead, service-heavy.** Offline TA/TO over-represented (38%), special requests +109%, total nights +14%, lead time -23%; 31% cancellation, highest ADR (115€). *Higher-touch channel, intermediate risk.*
- **C3: Non-refundable groups, very long lead.** Groups (58%) + Non Refund (63%); lead time +91% (median 260 days), previous cancellations +1,685%, repeated-guest +448%, low ADR (69€), virtually no booking changes; 99.2% cancellation. Speculative group blocks: strong overbooking/forecast signal.

The K-Means-Ward cross-tabulation confirms that Ward at K=2 merges C0 + C1 + C2 into one cluster and only isolates C3, strictly less informative for downstream decisions.

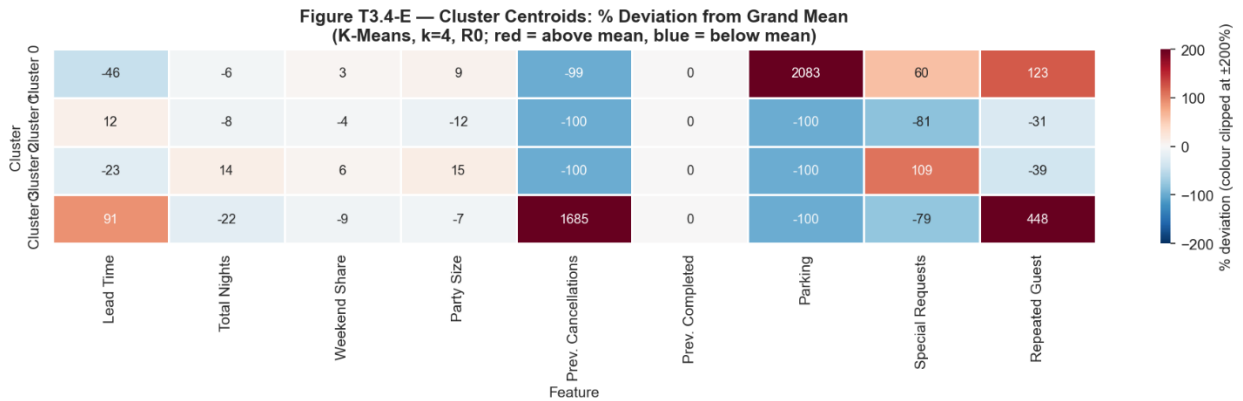


Figure 2| Per-feature deviation from the grand mean (%) for the four K-Means clusters on R0. Each row is a cluster; each column an input feature. The signature of C3 (lead time, previous cancellations, repeated guest) and of C0 (parking, special requests) is visually distinct.

5.4 Soft-clustering check (Fuzzy C-Means)

Fuzzy C-Means at K = 4 is applied to the same R0 representation as the partitional methods to test how cleanly the four behavioral segments separate under a soft-membership criterion. The fuzzifier m is calibrated on a {1.05, 1.08, 1.10, 1.13, 1.15, 1.20, 1.50, 2.00} grid (Appendix Table A8): $m \in [1.05, 1.10]$ yields near-crisp partitions (Partition Coefficient ≥ 0.89 , ARI vs. K-Means ≥ 0.85), $m \geq 1.50$ degenerates to a uniform membership matrix (PC = 0.25, ARI = 0.48), and the intermediate range sits in the "soft but informative" regime. $m = 1.13$ is selected as the smallest informative value: PC = 0.853, mean maximum membership = 0.898, ARI vs. K-Means = 0.827.

Membership confidence (Appendix Figure A3) is markedly asymmetric across clusters. C0 (parking-equipped resort families) is 100% confident at the max-membership ≥ 0.80 threshold and C3 (non-refundable groups) is 99.9%: every booking has a near-1 membership in its assigned cluster, confirming these two segments are structurally distinct. C1 (76.9%) and C2 (77.6%) carry a long left tail of bookings with split memberships, and 6–8% of cases lie below the 0.60 confidence threshold. The fuzzy structure is therefore concentrated entirely on the C1-C2 boundary, which matches both the bootstrap variability of K-Means in subsection 5.2 and the operational reading that "mainstream online transient" and "offline service-heavy" are adjacent rather than disjoint segments. Far from contradicting the K-Means solution, FCM confirms K = 4 and quantifies where the partition is honest about its own uncertainty.

5.5 Cluster-aware anomaly detection

To probe whether the K-Means centroids produce an actionable partition, every booking is ranked by its Euclidean distance to its assigned centroid in the standardized R0 space and the top 20 are inspected (Appendix Table A9). All twenty are assigned to cluster C2, the only one of the four clusters whose centroid is not anchored by a strong categorical signature (C0 by parking, C3 by previous cancellations, C1 by the Online TA $\times$ city-hotel volume), making C2 the natural absorber of off-distribution records.

Two qualitative types appear. Five bookings labelled *data-quality* (DQ) show party_size z-scores around $+37\sigma$: 26-person "transient" reservations at the Resort Hotel with ADR = 0 and identical 7-night stays — i.e. operational data-entry errors that should be removed from any future supervised training set. The remaining fifteen *rare-but-plausible* (RBP) bookings are genuine long stays of 35–56 nights with party_size ≈ 2 and non-zero ADR, dispersed across Offline TA/TO, Online TA and Groups market segments. These are real bookings that the centroid cannot fully represent but that nevertheless belong to the long-stay axis of C2, reinforcing rather than undermining its identity. The exercise validates that the partition is not masking a missing fifth cluster: the most atypical bookings are individually rare, not collectively a coherent segment.

5.6 Linear dimensionality reduction (PCA)

A controlled representation variant projects R0 onto its top principal components and refits K-Means in the reduced subspace, testing whether the segmentation depends on the full 57-dimensional one-hot expansion or survives compression. The scree and cumulative-variance plots (Appendix Figure A5) decay smoothly with no clear elbow; reaching 80% / 90% / 95% explained-variance ratio (EVR) requires 11 / 21 / 28 components respectively. The PCA loadings show that PC1 already contrasts last-minute special-requests-heavy bookings (Online TA, party_size, special_requests) against long-lead repeat-cancellers (lead_time, previous_cancellations), i.e. the dominant C0-C3 contrast is captured by the first component, suggesting most of the cluster signal lives in the early PCs.

The EVR sweep (Appendix Table A10) confirms this. At EVR = 90% (21 components) the best Silhouette in the PCA subspace reaches 0.165 at $k = 8$ (Table A10); at $k = 4$ — the value used for R0 — it is 0.157, with ARI = 0.93 against the K-Means R0 partition: same cluster identities, marginally improved compactness, consistent with mild denoising. At $\text{EVR} \leq 0.80$ the partition collapses to $k = 2$ (ARI ≈ 0.04 vs. R0), because dropping too many one-hot directions discards the categorical signal that distinguishes C1 from C2. PCA is therefore a benign representation choice rather than a substantive improvement: it does not change the segmentation, and we retain R0 as primary because interpretability of original-feature centroids (Appendix Table A11) is more valuable than the marginal Silhouette gain.

6 Discussion

Which result best answers the segmentation question. K-Means at $K = 4$ on R0: tied-best internal indices, ARI 0.995 across seeds, cross-method confirmation against iK-Means (ARI 0.994) and PCA (ARI 0.93). Ward's higher Silhouette reflects granularity, not quality, its $K = 2$ partition hides three of the four behavioral segments and cannot support the operational distinctions C0/C1/C2 imply.

Influence of preprocessing and similarity choices. Two choices dominate. Index-time discipline, excluding is_canceled, booking_changes, assigned_room_type and reservation_status from inputs makes the 99% cancellation alignment of C3 a genuine finding rather than a tautology: C3 is recovered purely from variables known at booking time. Scaler choice matters: RobustScaler (R1) compresses the long-tail signal in lead_time and previous_cancellations that separates C3 from the rest (ARI vs R0 = 0.18), so it is rejected; PCA-21d is benign (ARI 0.93) and modestly improves Silhouette.

Coherence, stability, interpretability. Each cluster has a distinctive numerical fingerprint and a dominant categorical mode. Seed-stability is good; bootstrap variability concentrates on the C1-C2 boundary, which FCM independently confirms (6–8% ambiguous bookings). Each cluster maps to a concrete operational lever: overbooking/forecast adjustment for C3, amenity capacity for C0, OTA contract and deposit review for C1, service-SLA design for C2.

What should not be concluded. There is no ground truth, so we cannot claim these are the "true" segments, only the most reproducible behavioral decomposition of the data support at $K = 4$. Silhouette ≈ 0.14 reflects the high-dimensional one-hot representation and should not be read as "weak clusters" without that context. The C3 cancellation signal is descriptive of risk.

7 Limitations, Ethics, and Reproducibility

Limitations. (i) No ground-truth labels; validity is internal + post-hoc. (ii) Restriction to TA/TO excludes Direct and Corporate channels: results do not generalise to those. (iii) Two hotels, both Portugal-dominant, 2015–2017; external validity to other markets and periods is untested. (iv) High-dimensional one-hot encoding caps the achievable Silhouette; mixed-type metrics (Gower, learned embeddings) are future work. (v) The C1-C2 boundary is genuinely fuzzy: a soft-classification deployment is more honest than a hard label for bookings near that boundary.

Reproducibility. Repository tag https://github.com/Leonorafonso28/hotel_booking_demand; entry point run_all.py; pinned environment in environment.yml (Python 3.11, scikit-learn 1.4, scikit-fuzzy). Every run is logged in experiments.csv with date, run_id, representation_id, method, parameters, seed, sample_rule, K, indices and runtime; all figures and tables regenerate from /results.

Ethics and governance. The dataset is anonymised and openly published; we comply with non-commercial academic terms and do not commit the raw file. country is handled only at aggregate level (rare-grouped at 1% + "Other"), so no individual-level inference is attempted. The analysis is descriptive, not deployed as an automated decision system.

Academic integrity. All code is original to the team; standard libraries are cited where used; declarations are included per course rules.

8 Conclusion

K-Means at $K = 4$ on representation R0 yields a small, coherent, stable and interpretable segmentation of the TA/TO booking population: parking-equipped resort families (C0), mainstream online transient city bookers (C1),

offline TA/TO long-lead service-heavy bookers (C2), and non-refundable group blocks with 99% cancellation (C3). The solution is robust across seeds, against iK-Means as a partitional alternative, and against PCA as a representation variant. The natural next step is supervised validation of C3 as a cancellation-risk indicator on out-of-time data, and soft-classification deployment on the fuzzy C1-C2 boundary.

References

1. António, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41-49. <https://doi.org/10.1016/j.dib.2018.11.126>
2. Course Materials (Slides and Labs)

Appendix

Table A1| K-Means (k-means++, n_init=5): full k-grid results (R0, Euclidean, seed = 12345)

k	Inertia	Silhouette $\uparrow$	Calinski-Harabasz $\uparrow$	Davies-Bouldin $\downarrow$	Time (s)
2	943,259.1	0.1162	11,606	2.6227	0.38
3	847,131.9	0.1288	12,007	2.2530	0.42
4	755,066.0	0.1403	12,954	1.9220	0.50
5	705,990.4	0.1382	12,089	1.8963	0.39
6	657,885.2	0.1391	11,808	1.8346	0.67
7	619,103.8	0.1346	11,477	1.7971	0.51
8	586,048.7	0.1425	11,179	1.7292	0.53

Table A2| iK-Means (AP initialisation, n_init=1): full k-grid results (R0, Euclidean, seed = 12345)

k	Inertia	Silhouette $\uparrow$	Calinski-Harabasz $\uparrow$	Davies-Bouldin $\downarrow$	Time (s)
2	943,259.6	0.1162	11,606	2.6235	0.07
3	847,132.2	0.1288	12,007	2.2530	0.13
4	755,063.3	0.1403	12,954	1.9215	0.12
5	705,990.4	0.1382	12,089	1.8963	0.16
6	706,364.0	0.1270	9,656	1.9907	0.16
7	649,860.5	0.1156	10,162	2.1907	0.10
8	610,590.1	0.1179	10,169	1.9783	0.11

Table A3| Ward: full k-grid results (R0, Euclidean, seed = 12345)

k	Silhouette	Cal-Har	Davies-B	Time (s)
2	0.4124	9,317	1.6349	2.74
3	0.2871	9,655	1.4963	2.91
4	0.2955	9,794	1.1616	2.70
5	0.1197	10,379	1.7094	2.93
6	0.1232	10,302	1.8436	2.76

Table A4 | Controlled Preprocessing Sensitivity Check

Representation	Scaler	Log transform	Silhouette	Calinski-Harabasz	Davies-Bouldin	ARI vs. R0
\$R_0\$(Baseline)	StandardScaler	No	0.1403	12,956.3	1.9212	1.0000
\$R_1\$(Robust)	RobustScaler	No	0.1262	11,931.4	2.2232	0.1756

Table A5 | Statistical Stability and Verification Matrix ($\mu \pm \sigma$) across Seeds/Resamples

Method	K	Silhouette (mean $\pm$ std)	CH (mean $\pm$ std)	DB (mean $\pm$ std)	Pairwise ARI (mean $\pm$ std)
K-Means	4	0.1394 $\pm$ 0.0012	12,956.2 $\pm$ 0.1	1.9215 $\pm$ 0.0004	0.9947 $\pm$ 0.0034
iK-Means	4	0.1308 $\pm$ 0.0151	12,280.7 $\pm$ 1,088.1	2.0197 $\pm$ 0.1599	0.7108 $\pm$ 0.2893
Ward	2	0.3161 $\pm$ 0.0018	11,961.5 $\pm$ 0.0	1.1512 $\pm$ 0.0000	1.0000 $\pm$ 0.0000

Table A6| Cross-method pairwise Adjusted Rand Index between the three clustering methods at their selected K (K-Means K=4, iK-Means K=4, Ward K=2; representation R0, seed = 12345).

	K-Means (K=4)	iK-Means (K=4)	Ward (K=2)
K-Means (K=4)	1.0000	0.9937	0.1549
iK-Means (K=4)	0.9937	1.0000	0.1548
Ward (K=2)	0.1549	0.1548	1.0000

Table A7| Row-normalised contingency between the K-Means K=4 partition and the Ward K=2 partition

K-Means cluster	Ward 0	Ward 1
C0	0.9993	0.0007
C1	1.0000	0.0000
C2	1.0000	0.0000
C3	0.0000	1.0000

Table A8| Fuzzy C-Means fuzzifier m calibration at K=4 on R0 (seed = 12345). PC = Partition Coefficient

m	PC $\uparrow$	mean max w	min proto dist	n_iter	ARI vs K-Means	Verdict
1.05	0.9501	0.9654	1.9615	13	0.9431	near-crisp
1.08	0.9133	0.9395	1.9306	18	0.8743	near-crisp
1.10	0.8894	0.9229	1.9063	17	0.8484	near-crisp
1.13	0.8535	0.8978	1.8607	15	0.8265	soft, informative
1.15	0.8291	0.8806	1.8234	14	0.8179	soft, informative
1.20	0.7661	0.8353	1.7041	12	0.8067	soft, informative
1.50	0.2500	0.2509	0.0008	50	0.4841	DEGENERATE
2.00	0.2500	0.2501	0.0001	14	0.4942	DEGENERATE

Table A9| Top-20 most atypical TA/TO bookings ranked by anomaly score (standardised distance from the assigned K-Means centroid).

#	Score	Type	Lead	Nights	Party	Market segment	Canc.	ADR (€)	Hotel
1	13.03	DQ	361	7	26	Offline TA/TO	1	0.0	Resort
2	13.03	DQ	354	7	26	Offline TA/TO	1	0.0	Resort
3	13.03	DQ	347	7	26	Offline TA/TO	1	0.0	Resort
4	13.02	DQ	340	7	26	Offline TA/TO	1	0.0	Resort
5	13.02	DQ	333	7	26	Offline TA/TO	1	0.0	Resort
6	7.35	RBP	322	56	2	Offline TA/TO	1	25.5	Resort
7	7.30	RBP	71	56	2	Online TA	0	28.8	Resort
8	6.20	RBP	206	48	2	Online TA	0	0.0	City
9	5.90	RBP	30	46	2	Online TA	0	0.0	Resort
10	5.55	DQ	55	14	12	Offline TA/TO	1	133.2	Resort
11	5.40	RBP	214	42	2	Offline TA/TO	0	31.4	Resort
12	5.39	RBP	150	42	2	Offline TA/TO	0	39.4	Resort
13	5.38	RBP	242	42	2	Offline TA/TO	0	35.7	Resort
14	5.38	RBP	122	42	2	Offline TA/TO	1	22.1	Resort
15	5.35	DQ	37	2	12	Online TA	0	84.5	City
16	4.87	RBP	80	38	2	Groups	0	0.3	Resort
17	4.51	RBP	92	35	1	Offline TA/TO	0	36.9	Resort
18	4.46	RBP	148	35	1	Offline TA/TO	0	42.0	Resort
19	4.43	RBP	192	35	2	Offline TA/TO	0	34.0	Resort
20	4.40	RBP	140	35	2	Offline TA/TO	0	45.0	Resort

Table A10| K-Means partitions in PCA-reduced subspaces (R0 to PCA at EVR thresholds from 50% to 95%)

EVR threshold	n components	best K	best Silhouette ↑	ARI vs. R0
0.50	5	6	0.2917	0.5308
0.70	8	2	0.4157	0.0433
0.80	11	2	0.3833	0.0433
0.90	21	8	0.1650	0.9253
0.95	28	8	0.1533	0.9214

Table A11| K-Means K=4 cluster centroids back-projected to original-feature units (median of post-cap values, R0 inputs).

Cluster	Lead time	Prev. canc.	Prev. compl.	Total nights	Weekend share	Party size	Parking	Special req.
Cluster 0	62.7	0.00	0.00	3.32	0.27	2.19	1.01	0.97
Cluster 1	88.1	0.00	0.00	4.02	0.28	2.31	0.00	1.26
Cluster 2	129.4	0.00	0.00	3.26	0.25	1.77	0.00	0.11
Cluster 3	220.6	1.00	0.00	2.77	0.23	1.87	0.00	0.13

Figure A1| Ward agglomerative clustering on the R0 representation (10,000-row stratified subsample, seed = 12345).

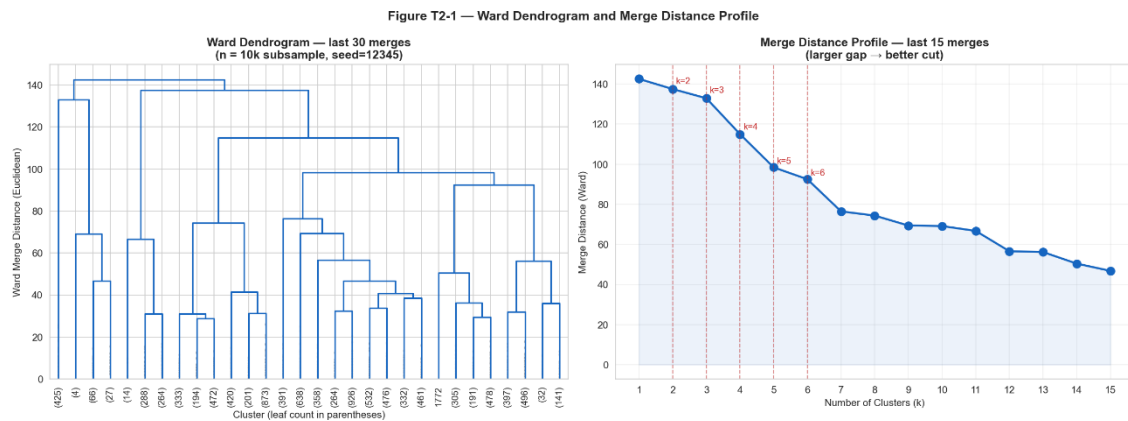


Figure A2| Radar profile of the four K-Means clusters (k = 4 on R0)

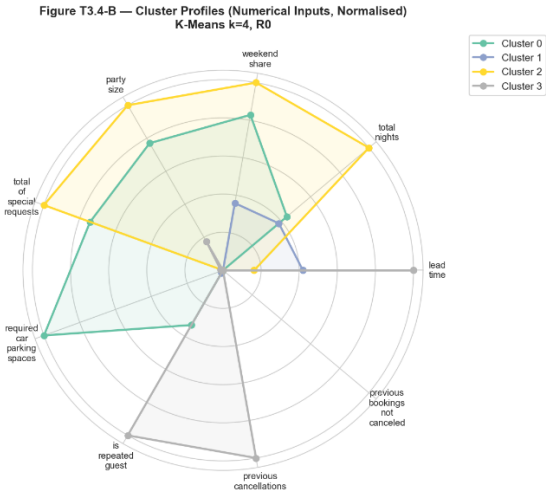


Figure A3| Fuzzy C-Means membership confidence at the calibrated fuzzifier $m = 1.13$, $k = 4$ on R0.

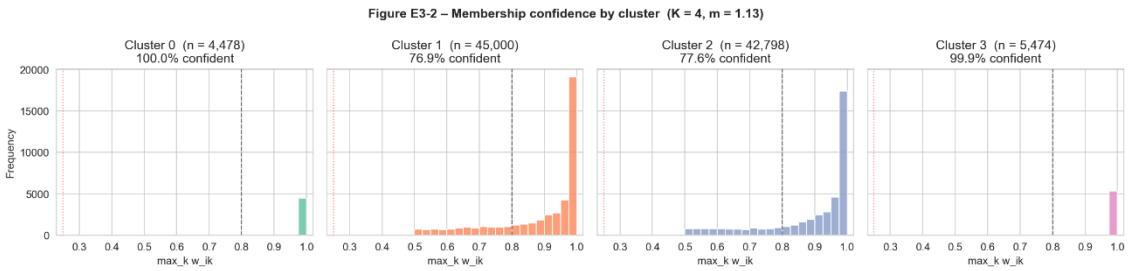


Figure A4| Z-score heatmap of the Top-20 most atypical TA/TO bookings

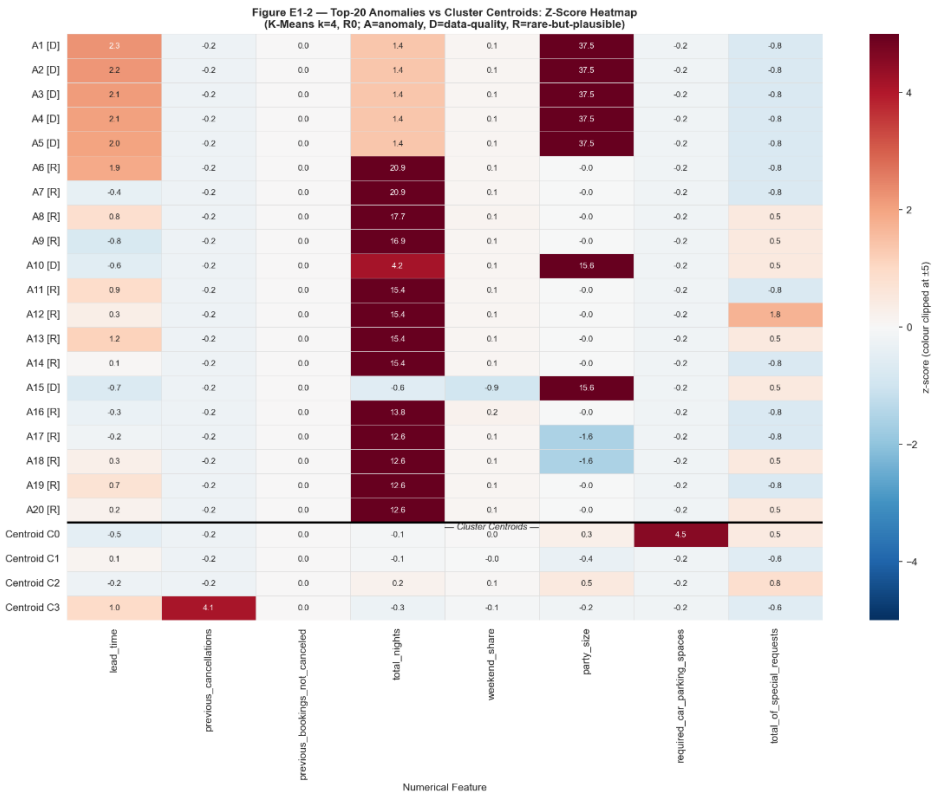


Figure A5 – Scree Plot and Cumulative Explained Variance

